

MAHIR PATEL

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EDUCATION

University of Illinois Urbana-Champaign — Grainger College of Engineering

Aug 2025 – May 2029

B.S. Electrical Engineering (intended) · Minors: Computer Science, Mathematics · GPA: 3.70 / 4.00 · Dean's List

Coursework: Data Structures & Algorithms, Object-Oriented Programming (C++), Computer Systems (ECE 120), Electronics (ECE 110), Linear Algebra, Differential Equations

Certifications: NVIDIA Deep Learning Institute · FCC Amateur Radio Technician License

EXPERIENCE

Co-Founder & Engineer — Ringlatch

Champaign, IL · Aug 2026 – Present

- Built and launched an AI phone receptionist for local businesses — answers after-hours calls, books owner-confirmed appointments, bilingual English/Spanish, emails owners a summary of every call; live with a personalized demo line.
- Own the entire stack solo: Retell voice agents, Cloudflare Workers backend, Supabase/Postgres, Stripe billing, and automated per-client provisioning from a 7-question intake — a new client can go live the same day.

Technical Coordinator — IEEE UIUC Student Branch

Aug 2025 – Present

- Lead all technical programming for the branch — run hands-on workshops in KiCad PCB design, microcontroller/embedded labs, and soldering, and organize the technical build projects and event hardware.
- Mentor members on embedded and hardware projects end-to-end, from schematic and PCB layout through firmware bring-up and debugging on the bench.

Technical Vice President, SIGAIDA — ACM @ UIUC

Aug 2025 – Present

- Serve on the executive board and lead technical programming — hands-on AI/data workshops, member projects, and speaker sessions — for ACM@UIUC's AI & data special interest group.

Electrical Team Lead — ASME at Illinois

Champaign, IL · Oct 2025 – Apr 2026

- Led the electrical subsystem for a team-built Iron Man-style animatronic helmet, integrating a Raspberry Pi 5, servo-actuated faceplate, and eye-tracking LED arrays across a 6-person team.
- Programmed a Python "JARVIS" voice assistant (speech-to-text + LLM API) for hands-free control of servos and lighting, achieving sub-second response.

Undergraduate Researcher — URSA (Magnet: LLM Serving Systems)

Urbana, IL · Aug 2025 – Jan 2026

- Researched Magnet, a serving system that treats multi-agent LLM pipelines as dataflow graphs — overlapping downstream context prefill with upstream token generation via just-in-time and parallel prefilling.
- Instrumented multi-agent workloads to collect execution traces (token rates, prefill times, KV-cache memory, dependencies); built a trace-driven simulator for just-in-time prefill scheduling and evaluated latency, throughput, and GPU utilization vs. sequential baselines.

PROJECTS

AI Driving Assistant

Python, PyTorch, Qwen3-VL, REST APIs, Vector DB

- Fine-tuned a Qwen3-VL vision-language model to grade driver performance frame-by-frame, evaluated on public dashcam datasets at 24-60 fps, returning actionable feedback with visual references.
- Built the full backend — REST APIs and a vector database for low-latency retrieval — serving model inference to a mobile app.

GestureLab — Gesture-Controlled Desk Assistant

Embedded C, React, Three.js, LLM APIs

- Built a dual-board embedded system integrating 10 sensor modules with a React/Three.js dashboard, Spotify control, and an LLM-powered focus coach — delivered end-to-end in a 24-hour hackathon.

NOVA — AI Assistant for UIUC Students

Python, LLM APIs, Retrieval, Discord

- Built and launched a retrieval-augmented assistant answering course and campus questions, deployed on Discord and used by 100+ UIUC students, iterating on live user feedback.

32-Bit ALU

SystemVerilog, Xilinx Vivado, RTL Design, Verification

- Designed and verified a 32-bit ALU with a self-checking Vivado testbench — directed and randomized vectors covering all operations and flag conditions.

Custom PCB Data Logger

KiCad, Embedded C, I2C/SPI

- Designed, routed, and hand-assembled a microcontroller data logger integrating 512 KB EEPROM and a real-time clock over I2C; validated signal integrity across 3 board revisions with an oscilloscope and logic analyzer.

TECHNICAL SKILLS

Languages: Python, C, C++, SystemVerilog, JavaScript, MATLAB, HTML/CSS

Hardware & EDA: KiCad, PCB layout, Xilinx Vivado, LTspice, RTL design & verification, oscilloscope, logic analyzer, Raspberry Pi, Arduino, STM32

ML & Software: PyTorch, LLM APIs, vector databases, REST APIs, React, Three.js, Node.js, Git, Linux